

ARARAT, Australia

WMO: 948340

Lat: 37.277S

Lon: 142.981E

Elev: 971

StdP: 14.19

Time Zone: 10.00 (AUS)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.9	32.7	25.7	20.4	49.1	28.7	23.5	48.1	30.0	48.2	29.2	49.4	1.4	200	0.556

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	25.0	93.2	63.5	89.3	62.8	85.5	61.6	67.9	81.8	66.0	81.8	64.3	78.6	10.9	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.1	92.8	71.6	61.5	84.7	69.3	59.1	77.4	67.9	32.8	82.2	31.2	81.9	30.0	78.8	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.1	29.1	23.4	25.5	102.0	1.7	3.0	24.3	104.1	23.3	105.8	22.3	107.5	21.1	109.7		
(5)			24.7	71.0	1.8	2.0	23.4	72.4	22.4	73.6	21.4	74.7	20.1	76.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.2	66.3	67.2	61.7	55.1	50.5	47.4	46.4	47.1	50.2	52.8	57.5	61.8		
	DBStd	9.15	7.29	7.36	6.39	5.57	4.45	4.08	3.35	4.08	5.20	5.31	6.47	7.43		
	HDD50	474	0	0	0	13	47	97	119	109	57	26	6	0		
	HDD65	3914	72	55	145	304	449	529	578	556	447	381	243	155		
	CDD50	2390	504	482	363	165	63	18	6	19	62	111	230	367		
	CDD65	353	112	117	42	5	0	0	0	0	1	2	17	57		
	CDH74	5279	1608	1540	602	66	2	0	0	0	4	66	315	1076		
	CDH80	2233	736	698	200	8	0	0	0	0	0	14	91	486		
(14) Wind	WSAvg	8.9	11.1	10.0	8.3	7.3	6.3	7.8	7.5	8.1	9.3	9.5	10.5	11.0		
(15) Precipitation	PrecAvg	23.4	1.5	1.3	1.3	1.7	2.1	2.3	2.5	2.8	2.5	2.4	1.7	1.4		
	PrecMax	35.8	5.4	8.9	4.0	4.3	5.7	5.0	4.5	5.3	4.6	8.0	3.0	4.0		
	PrecMin	11.9	0.1	0.0	0.2	0.2	0.3	0.4	0.7	0.5	0.7	0.5	0.4	0.0		
	PrecStd	4.8	1.1	1.7	0.9	1.3	1.2	1.2	1.1	1.1	1.2	1.5	0.7	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.1	98.7	90.0	79.8	71.1	61.3	59.1	65.8	73.3	81.9	88.5	96.4		
		MCWB	66.0	66.3	63.1	57.6	54.8	53.4	50.7	55.3	57.6	57.8	60.4	62.6		
	2%	DB	92.8	93.0	85.2	75.6	66.5	58.0	55.9	60.6	67.4	74.1	83.5	90.7		
		MCWB	63.8	64.4	61.7	57.6	53.4	52.7	48.9	50.8	55.6	56.5	59.3	61.0		
	5%	DB	88.2	88.4	80.8	71.5	62.3	55.8	53.8	57.5	63.7	69.2	77.5	85.9		
		MCWB	63.3	63.9	60.5	56.4	52.8	50.8	48.2	49.9	54.1	55.1	58.3	60.1		
	10%	DB	83.5	83.7	76.3	67.5	58.9	53.8	52.2	54.9	60.1	64.5	72.5	80.1		
		MCWB	61.8	62.7	59.3	55.5	52.1	49.6	47.5	48.6	52.3	54.0	56.7	59.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.5	71.5	67.4	61.2	58.1	56.9	52.2	57.6	61.0	60.9	65.7	66.9		
		MCDB	83.7	87.0	82.3	69.7	63.5	58.2	55.8	63.6	68.8	72.1	77.1	77.7		
	2%	WB	67.0	69.1	64.4	59.4	56.0	53.4	50.4	52.9	57.7	58.4	62.8	64.5		
		MCDB	86.6	79.8	77.8	70.2	61.4	56.7	53.6	57.5	65.6	68.0	75.4	77.8		
	5%	WB	65.2	67.0	62.0	58.0	54.4	51.6	49.3	50.8	55.0	56.7	60.5	62.4		
		MCDB	82.0	80.3	74.8	68.7	60.1	55.0	52.4	56.1	61.1	66.7	73.0	78.6		
	10%	WB	63.5	65.1	60.3	56.2	53.0	50.1	48.3	49.2	52.8	54.7	58.1	60.7		
		MCDB	78.6	80.0	74.1	65.8	58.6	52.9	51.4	53.9	59.0	64.0	69.3	77.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	25.7	25.0	22.7	20.6	16.6	12.1	11.7	13.9	16.2	19.1	22.0	25.3		
		MCDBR	36.0	34.1	32.2	28.7	24.1	14.3	13.9	18.7	21.5	29.6	33.0	38.5		
		MCWBR	11.7	10.7	12.3	13.4	13.5	9.4	8.8	11.2	12.3	14.8	13.5	12.7		
	5% WB	MCDBR	28.4	24.9	25.1	23.4	19.5	10.7	12.3	16.8	18.9	24.0	24.3	28.4		
		MCWBR	11.0	9.9	10.8	12.3	11.9	7.7	8.5	10.9	12.2	13.1	12.2	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.341	0.331	0.316	0.307	0.297	0.288	0.287	0.300	0.325	0.329	0.333	0.328		
	taud		2.473	2.520	2.554	2.559	2.564	2.583	2.582	2.526	2.441	2.445	2.470	2.506		
	Ebn at Noon		315	311	303	287	271	265	272	285	295	307	315	320		
	Edh at Noon		37	34	30	26	23	21	22	26	33	36	37	36		
(44) All-Sky Solar Radiation	RadAvg		2365	2073	1636	1142	763	605	660	935	1304	1734	2052	2303		
	RadStd		137	102	85	70	38	49	41	61	90	143	136	126		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	+0.86	N/A	N/A	+1.58	+0.74	N/A	-23	-229	+304	+97

Nomenclature: See separate page